Kushagra Katiyar

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Education

University of Florida Gainesville, FL

Combined Bachelor's and Master's in Computer Science, Minor in Statistics | GPA: 3.92 *Expected May 2028*

- **Certifications**: Introduction to Deep Learning (Nvidia), AI Certificate (University of Florida)
- **Coursework:** Data Structures and Algorithms, Operating Systems, Database Management, Programming Language Principles, Computer Architecture, Artificial Intelligence Fundamentals, UX Design, Linear Regression, Categorical Analysis

Professional Experience

Software Engineering Research Assistant — University of Florida Grasshopper Lab September 2026 – Present

- Architecting a **distributed, multi-agent simulation platform** letting professors spin up "experiences" where students onboard to a codebase alongside a **PM agent**, and specialized **teammate agents**, integrated with GitHub issues.
- Building on a **Python, Go, React, Kubernetes, Next.js, MongoDB** stack on **AWS/Cloudflare** with **serverless**, event-driven agent orchestration, benchmarking open-source LLMs for cost-efficiency.

Primo Brands - Software Engineering Intern (AI & Digital Experience) August 2026 – Present

- Engineered a **data ingestion pipeline** to collect organization wide SAP financials, **50+ Power BI** and **50+ Qualtrics** exports via **REST APIs** into **Azure Blob** storage feeding quarterly AI report generation for **investor relations calls**.
- Automated quarterly report generation atop the consolidated dataset with agentic validation, cutting prep time by **15 days** and replacing manual cross-referencing across **400,000+** data points and pages of Excel exports.

Primo Brands - Software Engineering Intern (AI & Digital Experience) May 2026 – July 2026

- Shipped a chatbot serving **12,000+** employees backed by a hybrid **BM25 + vector + semantic search** index grounded in **72+** ServiceNow articles and **100,000+** tickets, resulting in a **23% reduction** in IT helpdesk calls.
- Designed an **LRU semantic caching layer** for user inputs that cut response latency **71% (7s to 2s)** and reduces redundant LLM inference calls with a **70%** cache hit rate based on query semantics rather than direct string values.
- Designed the **FastAPI + React** system architecture end-to-end, optimizing image preprocessing (Doc Intelligence Layouts) to cut per-article inference costs **10%**, and presenting trade-offs to the cloud infrastructure manager.
- Built a citation-validation safeguard that verifies every referenced image against source data in **Azure Blob** storage before rendering it in-chat, preventing employees from acting on fabricated instructions.

Software Engineering Research Assistant — University of Florida Virtual Learning Lab January 2026 – Present

- Built a concurrent evaluation pipeline (**24** parallel workers via **asyncio** and **Pydantic**-validated outputs) that sped up batch grading **500%**, powering QA for **Storiza**, a personalized children's reading platform reaching **200+** kids in Gainesville, FL.
- Designed an 8-metric rubric scored by an ensemble of independent evaluators to grade AI-generated K-3 phonics stories, improving evaluator-human agreement (Cohen's Kappa) from **0.4 to 0.72** and enabling faster rubric iteration.

Software Engineering Intern — English2Success February 2025 – July 2025

- Developed a **Flask** backend running **Cloud Run** jobs to automate **80+** weekly survey-summary reports, saving **8+ hours** of weekly manual overhead when providing english tutoring to kids across **7 countries**, enabled growth from **200 to 450** users.

Involvement

Quantitative Developer — AlgoGators September 2026 – Present

- Built a causal **Savitzky-Golay** trend-following signal with a **4-horizon momentum ensemble** and standard-deviation-based position risk sizing across **5 uncorrelated ETFs** for UF's quantitative trading student hedge fund managing **$500,000 AUM**.
- Caught a lookahead-bias bug that inflated the backtested Sharpe to **14**, correcting it to **0.60** by making the algorithm causal.
- Backtested the strategy: **+6.12%** vs. **-25.83%** buy-and-hold during the 2008 GFC, winning **15/19 (79%)** market crashes.

Software Developer – Dream Team Engineering January 2026 – Present

- Collaborating with a neurosurgeon resident at **UF Shands Hospital**, to design an Android alternative to **$3,000+** of clinical hardware to conduct Train-of-Four anesthesia testing, with upcoming **IRB-approved** trials in operating rooms.
- Implemented **MediaPipe** landmarks and **OpenCV** Lucas-Kanade optical flow with Savitzky-Golay smoothing in **Kotlin** and **Java**, to detect twitches while removing camera shake, **90%** accuracy at a **5%** rate of error across **400+** Unity videos.

Personal Projects

Buddy! Lock In – SaseHacks, **1st Place** Gamification & Best Use of Blender March 2026

- Architected an event-driven **Node.js** backend pushing real-time focus state, quiz events, and screen analysis across **4 concurrent players** over **Socket.io** rooms, keeping all 3D pet states and HUD in sync with **<100ms** event latency.
- Built a focus pipeline: **MediaPipe** face landmarks ran in-browser paired with 3D avatars reacting to focus state with animations and sounds, storing quizzes in **MongoDB** for reuse across sessions, rendering in **React Three Fiber**.

SteamSearch – Data Structures & Algorithms TA Favorite Nominee August 2025

- Built a C++ game recommendation engine computing **100,000+** games (cosine similarity, MinHash, Jaccard) in **200ms**, turning **600MB** of data into a **125MB** binary format with pooled strings to cut deployment memory from **1GB** to **300MB**.
- Ingested metadata through a web API with **20 EC2s** concurrently storing into an **S3 bucket** in **6 hours instead of 120**.

Technical Skills

Languages: Python, TypeScript/JavaScript, Kotlin, Go, Java, C/C++, SQL, HTML/CSS
AI & ML: LangGraph, Pydantic, OpenCV, MediaPipe, Pandas, scikit-learn, TensorFlow, RAG, Claude Code, Copilot, Codex
Backend & Fullstack: FastAPI, Flask, REST, WebSockets, PostgreSQL, MongoDB, React, Next.js, SpringBoot, Node.JS
Cloud & Infrastructure: AWS, Azure, GCP, Cloudflare, Docker, Kubernetes, CI/CD, Git, VS Code, Linux Scripting